

UNIVERSITY COLLEGE OF JAFFNA

UNIVERSITY OF VOCATIONAL TECHNOLOGY

HIGHER NATIONAL DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY

NATIONAL VOCATIONAL QUALIFICATION LEVEL 06

Initial Project Proposal
Online Food Ordering and Management System

Software Change Management (K72C001M12)

Assessment No 01

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JF/ICT/24/10

SOFTWARE REQUIREMENTS

SPECIFICATION (SRS)

Online Food Ordering & Management System

Requirements Baseline – Version 1.0

Document Information

Item	Details
Project	Online Food Ordering System
Organization	_____
Document	Software Requirements Specification
Version	1.0
Baseline	Requirements Baseline v1.0
Development Approach	Incremental Development with Controlled Change Management
Technology	HTML, CSS, Asp.Net C#, MySQL
Version Control	GitHub
Database	MySQL
Status	Proposed for Review
Prepared By	Student Project Team
Date	_____

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the Online Food Ordering System.

The document establishes the requirements that will be used as the reference for:

- System analysis
- System design
- Database design
- Software development
- Testing
- Release activities
- Software Change Management

The requirements in this document represent the initial proposed scope of the system. Once reviewed and approved by the relevant stakeholders, this document will become the Requirements Baseline v1.0.

Any new requirement introduced after Version 1.0 has been approved shall be handled through the Software Change Management process using a formal Change Request (CR).

1.2 Project Background

The project is an online food ordering website. The system is intended to allow customers to view available food items, select items, place an order, and provide the required customer information for the order. The system also provides administrative functions for managing food items and customer orders.

The Version 1.0 release is focused on the basic online food ordering process. Additional functionality may be considered for future releases through the Software Change Management process.

2. Problem Statement

A food ordering process that depends on manual communication can make it difficult for customers to see available food items and place orders in an organized way. Restaurant staff may also need to manage food information and orders manually.

The proposed system aims to provide a web-based method for viewing food items and submitting food orders, with the order information stored in a MySQL database.

3. Project Objectives

1. To develop a web-based online food ordering system.
2. To allow customers to view available food items.
3. To allow customers to select food items and place orders.
4. To store customer and order information in a MySQL database.
5. To allow the administrator to manage food items.
6. To allow the administrator to view and manage customer orders.
7. To reduce manual work in the food ordering process.
8. To provide a simple and understandable website interface.
9. To demonstrate controlled software changes using Software Change Management practices.

4. Project Scope

4.1 In Scope for Version 1.0

The Version 1.0 system shall provide:

Customer Functions

- Customer registration/login where required by the system
- View food items
- View food item details
- Select food items

- Add food items to an order
- Place an order
- Provide customer/contact information
- View order information/status where implemented
- Logout where authentication is provided

Administrator Functions

- Administrator login
- Add food items
- View food items
- Update food item information
- Delete food items
- View customer orders
- View order details
- Update order information/status where required
- Logout

Technical Scope

- HTML-based web pages
- CSS for website presentation
- Asp.Net C# for application/backend functionality
- MySQL database
- Web-based system
- Version control using GitHub

4.2 Out of Scope for Version 1.0

- Online payment gateway integration
- Advanced recommendation system
- Advanced analytics and reporting
- Loyalty/reward system
- Real-time chat
- Push notifications
- Multiple restaurant management
- Advanced role-based access control

These features may be introduced in future releases through approved Change Requests.

5. Stakeholders

5.1 Stakeholder Register

Stakeholder	Role	Interest / Responsibility
Administrator	System User	Manage food items and customer orders
Customer	End User	View food items and place food orders
Restaurant / Food Staff	Business User	Prepare and manage customer orders

Institute / Project Owner	Management	Monitor the system and future development
IT Administrator	Technical Stakeholder	Maintain and support the system
Project Team	Development Team	Analyze, design, develop, test, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Monitor project progress and SCM practices

6. User Roles

6.1 Administrator

The administrator is responsible for managing food items and customer orders.

- Log in.
- Add food items.
- View food items.
- View food item details.
- Update food item information.
- Delete food items.
- View customer orders.
- View order details.
- Manage order status where required.
- Log out.

6.2 Customer

The customer uses the system to view food items and place orders.

- Register or log in where required.
- View available food items.
- View food item details.
- Select food items.
- Place an order.
- Provide required customer information.
- View order information where provided.
- Log out where authentication is provided.

Customers shall not be permitted to manage food items or other customers' orders.

7. Functional Requirements

Functional requirements describe what the system shall do.

7.1 Authentication Requirements

FR-01 – User Login

The system shall allow an authorized user to log in using valid credentials where authentication is provided.

FR-02 – Login Validation

The system shall validate the credentials provided by the user.

FR-03 – Invalid Login Handling

The system shall display an appropriate error message when invalid login credentials are provided.

FR-04 – User Logout

The system shall allow an authenticated user to log out of the system.

7.2 Online Food Ordering Requirements

FR-05 – View Food Items

The system shall allow customers to view available food items.

FR-06 – View Food Details

The system shall display relevant information for a selected food item, such as name, price, and description where available.

FR-07 – Select Food Items

The system shall allow a customer to select food items for an order.

FR-08 – Create Order

The system shall allow a customer to submit an order containing selected food items.

FR-09 – Customer Information

The system shall collect the required customer information for an order.

FR-10 – Calculate Order Total

The system shall calculate the total amount of the selected food items based on the stored item prices.

FR-11 – Store Order

The system shall store valid order information in the MySQL database.

FR-12 – View Order

The system shall allow the customer to view relevant information about their submitted order where this function is provided.

FR-13 – Order Management

The system shall allow an authorized administrator to view customer orders.

FR-14 – Update Order Status

The system shall allow an authorized administrator to update the order status where order status management is implemented.

7.3 Online Food Ordering Profile Requirements

FR-15 – Customer Profile

The system shall maintain the customer information required for ordering.

FR-16 – View Own Information

An authenticated customer shall be able to view their own stored information where a customer profile is provided.

FR-17 – Restrict Customer Information

The system shall prevent a customer from accessing another customer's private information through normal system functions.

7.4 Food Management Requirements

FR-18 – Add Food Item

The system shall allow an authorized administrator to add a new food item.

FR-19 – Update Food Item

The system shall allow an authorized administrator to update food item information.

FR-20 – Delete Food Item

The system shall allow an authorized administrator to delete a food item.

FR-21 – View Food List

The system shall allow an authorized administrator to view the food item list.

8. Non-Functional Requirements

8.1 Security

NFR-01 – Password Security

User passwords shall not be stored as plain text where passwords are stored by the system.

NFR-02 – Authentication

Protected system functions shall require user authentication.

NFR-03 – Authorization

The system shall restrict administrator functions to authorized administrators.

8.2 Performance

NFR-04 – Response Time

Under normal operating conditions, commonly used pages and requests should respond within approximately three seconds.

NFR-05 – Database Performance

The database shall support efficient retrieval and modification of food, customer, and order records.

8.3 Usability

NFR-06 – User Interface

The system shall provide a simple, understandable, and consistent user interface.

NFR-07 – Validation Messages

The system shall provide clear messages when input is invalid or an operation cannot be completed.

8.4 Reliability

NFR-08 – Data Integrity

The system shall maintain the integrity and consistency of stored food, customer, and order information.

NFR-09 – Error Handling

The system shall handle expected application errors without causing the application to terminate unexpectedly.

8.5 Availability

NFR-10 – Availability

The system should be available during the normal operating period when deployed in the target environment.

8.6 Maintainability

NFR-11 – Maintainable Architecture

The system shall be organized so that future changes can be implemented with limited impact on existing functionality.

8.7 Portability

NFR-12 – Browser Compatibility

The web application should operate correctly on commonly used modern web browsers.

8.8 Responsiveness

NFR-13 – Responsive Interface

The user interface should adapt to common desktop, tablet, and mobile screen sizes where responsive design is included.

9. User Stories

US-01 – Login

As a registered user, I want to log in using my credentials so that I can access the functions available to me.

US-02 – View Food Items

As a customer, I want to view food items so that I can choose food to order.

US-03 – View Food Details

As a customer, I want to view food details so that I can understand the food item before ordering.

US-04 – Place Order

As a customer, I want to place an order so that the restaurant receives my food request.

US-05 – View Order

As a customer, I want to view my order information so that I can check what I ordered.

US-06 – Manage Food

As an administrator, I want to add, update, view, and delete food items so that the food menu is maintained.

US-07 – Manage Orders

As an administrator, I want to view and manage customer orders so that orders can be processed.

US-08 – Customer Profile

As a customer, I want to view my own information so that I can check the information used for ordering.

US-09 – Logout

As an authenticated user, I want to log out so that my account is protected when I finish using the system.

10. Acceptance Criteria

AC-01 – Login

- Given a registered user is on the login page, when valid credentials are entered and Login is selected, the system shall authenticate the user and display the appropriate page.
- If invalid credentials are entered, an appropriate error message shall be displayed.

AC-02 – View Food

- Given food items exist, when the customer opens the food/menu page, the system shall display the available food items.

AC-03 – Place Order

- Given the customer has selected valid food items and entered the required information, when the customer submits the order, the order shall be saved.
- The order total shall be calculated.
- A confirmation message shall be displayed.

AC-04 – View Order

- Given a valid customer order exists, when the customer opens the relevant order page, the system shall display the order information available to that customer.

AC-05 – Manage Food

- Given the administrator is authenticated, when valid food information is entered and saved, the food item shall be stored.
- The administrator shall be able to update and delete food items.

AC-06 – Manage Orders

- Given the administrator is authenticated, when the administrator opens order management, the system shall display available customer orders.
- The administrator shall be able to view order details and update order status where implemented.

AC-07 – Customer Information

- Given a customer is authenticated, when the customer opens the profile function, the system shall display the customer's own stored information.
- The customer shall not be able to access another customer's private information through normal system functions.

AC-08 – Logout

- Given a user is authenticated, when Logout is selected, the system shall terminate the authenticated session and return the user to the login page.

11. MoSCoW Prioritization

Priority	Requirements
Must Have	Food item viewing, food selection, order creation, customer information, order storage, administrator food management, administrator order viewing, basic authentication/security
Should Have	Order status viewing/updating, search/filter food items, improved validation, user feedback
Could Have	Basic dashboard, basic order history, basic categories
Won't Have in v1.0	Online payment gateway, GPS delivery tracking, mobile application, advanced recommendations, loyalty system, real-time chat, advanced analytics

12. Product Backlog

ID	Product Backlog Item	Priority	Related Requirements	Initial Status
US-01	User Login	Must	FR-01, FR-02, FR-03	Planned
US-09	User Logout	Must	FR-04	Planned
US-02	View Food Items	Must	FR-05	Planned
US-03	View Food Details	Must	FR-06	Planned
US-04	Place Order	Must	FR-07 to FR-11	Planned
US-05	View Order	Should	FR-12	Planned
US-06	Manage Food	Must	FR-18 to FR-21	Planned
US-07	Manage Orders	Must	FR-13, FR-14	Planned
US-08	View Customer Profile	Must	FR-15 to FR-17	Planned
—	Online Payment	Won't Have	—	Future Release
—	GPS Delivery Tracking	Won't Have	—	Future Release

13. Use Cases

13.1 Use Case List

Use Case ID	Use Case	Primary Actor
UC-01	Login	Administrator / Customer

UC-02	View Food Items	Customer
UC-03	View Food Details	Customer
UC-04	Place Order	Customer
UC-05	View Order	Customer
UC-06	Manage Food Items	Administrator
UC-07	View Customer Orders	Administrator
UC-08	Manage Order Status	Administrator
UC-09	View Own Profile	Customer
UC-10	Logout	Administrator / Customer

13.2 Use Case Example – Place Order

Field	Description
Use Case ID	UC-04
Name	Place Order
Actor	Customer
Preconditions	Customer can access the ordering function and required food items are available.
Trigger	Customer selects food items and chooses to place the order.
Main Flow	View food → Select food → Enter required information → Review order → Calculate total → Submit order → Save order → Display confirmation
Alternative Flow	Invalid or missing information; unavailable food item; order cannot be saved
Postcondition	Valid order information is stored successfully in the database.

14. Business Rules

BR-01

Only authenticated users may access protected system functionality.

BR-02

Only authorized administrators may add, update, or delete food items.

BR-03

Only authorized administrators may view or manage customer order information.

BR-04

Each food item shall have a unique system identifier.

BR-05

Each order shall have a unique order identifier.

BR-06

Food item prices used for an order shall come from the stored food item information.

BR-07

Required customer and order information must be provided before an order can be saved.

BR-08

Order total shall be based on the selected item quantities and prices.

BR-09

Customer private information shall not be accessible to other customers through normal system functions.

BR-10

Passwords must not be stored in plain text where passwords are stored by the system.

15. Data Requirements

Version 1.0 requires the system to maintain the following core information.

15.1 User Information

- User ID
- Username
- Password
- Role

15.2 Customer Information

- Customer ID
- First Name
- Last Name
- Email
- Phone Number
- Address where required for the order

15.3 Food Information

- Food ID
- Food Name
- Description
- Price
- Category where used
- Availability

15.4 Order Information

- Order ID
- Customer ID
- Order Date/Time
- Selected Food Items
- Quantity
- Unit Price
- Total Amount

- Order Status where implemented

Detailed database structures, relationships, normalization, constraints, and SQL schemas will be documented separately during the Database Design phase.

16. System Constraints

10. The application will be developed as a web-based system.
11. HTML will be used for web page structure.
12. CSS will be used for web page presentation.
13. Asp.Net C# will be used for application/backend functionality.
14. MySQL will be used as the database.
15. GitHub will be used for version control and SCM activities.
16. The Version 1.0 system will focus on essential online food ordering functionality.
17. New requirements after baseline approval must follow the Change Request process.

17. Assumptions

18. The project owner will provide the required food and order information for testing.
19. Users will have access to a modern web browser.
20. Required administrator credentials will be provided.
21. Food prices and availability will be maintained by authorized users.
22. Customer information entered into the system will be provided by the customer or authorized staff.
23. The final deployment environment will be defined at a later stage.
24. Future functionality can be introduced through controlled software changes.
25. Requirements may evolve after Version 1.0 and will be managed through the Software Change Management process.

18. Requirements Traceability Matrix

Requirement	User Story	Acceptance Criteria	Test Case
FR-01	US-01	AC-01	TC-01
FR-04	US-09	AC-08	TC-02
FR-05	US-02	AC-02	TC-03
FR-06	US-03	AC-02	TC-04
FR-07 to FR-11	US-04	AC-03	TC-05
FR-12	US-05	AC-04	TC-06
FR-13 to FR-14	US-07	AC-06	TC-07
FR-18 to FR-21	US-06	AC-05	TC-08
FR-15 to FR-17	US-08	AC-07	TC-09

The test case identifiers will be formally defined and expanded during the Test Planning and Testing sessions.

19. Requirements Review

Review Question 1

Can customers add, update, or delete food items?

Decision: No. Food item management is restricted to authorized administrators.

Review Question 2

Can customers manage other customers' orders?

Decision: No. Customers may only access their own permitted order information.

Review Question 3

Is online payment included in Version 1.0?

Decision: No. Online payment is outside the Version 1.0 baseline.

Review Question 4

Is GPS delivery tracking included?

Decision: No. GPS delivery tracking is outside the Version 1.0 baseline.

Review Question 5

Is a mobile application included?

Decision: No. The Version 1.0 project is a web-based system.

Review Question 6

Can customers modify administrator-managed food information?

Decision: No. Food information is managed by authorized administrators.

20. Version 1.0 Requirements Baseline

After stakeholder review and approval, the requirements contained in this document shall become the Requirements Baseline v1.0.

The baseline establishes the official reference point for Version 1.0 development.

After baselining:

- New requirements shall not be added directly to the Version 1.0 scope.
- Any new requirement must be submitted through the Software Change Management process.
- Approved changes shall be assessed for their impact on requirements, design, database, implementation, testing, and documentation.
- Relevant baseline and version information shall be updated when an approved change is implemented.

The general change process shall be:

New Requirement → Change Request → Impact Analysis → Change Approval → Implementation → Testing → Code Review → Merge → Documentation Update → New Version / Release

21. Configuration and Version Control

The approved requirements document shall be maintained under Git version control.

21.1 Baseline Information

Item	Value
Baseline Name	Requirements Baseline v1.0
Version	1.0
Status	Proposed for Review / Approved after Sign-off
Git Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____

The baseline tag requirements-v1.0 will provide a fixed reference point for future change analysis.

22. Future Change Management

Once the Version 1.0 requirements have been baselined, future requirements may be introduced through the formal change management process.

Example Future Request

"The business now requires online payment functionality."

This requirement shall not be automatically added to Version 1.0. Instead, the request shall follow the formal change process:

Customer / Business Request → CR-001 → Impact Analysis → CCB / Project Approval → Requirements Update → Design Update → Database Update → Backend Update → Frontend Update → Testing → Release v1.1

This process allows the project to demonstrate the Software Change Management lifecycle.

23. Version History

Version	Date	Description	Author	Status
0.1	_____	Initial requirements draft	_____	Draft
0.2	_____	Requirements reviewed and revised	_____	Review
1.0	_____	Requirements baseline established	_____	Approved

24. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined for Version 1.0 represent the agreed initial scope of the Online Food Ordering System.

Role	Name	Signature	Date
Client / Business Representative	_____	_____	_____
Project Team Representative	_____	_____	_____
Lecturer / Supervisor	_____	_____	_____

Appendix A – Version 1.0 Scope Summary

Included

- Authentication where required
- Food item viewing
- Food item details
- Food selection
- Order creation
- Customer information
- Order storage
- Administrator food management
- Administrator order management
- Basic security
- Basic access control
- Web interface
- C# application/backend
- MySQL database

Not Included

- Online payment gateway
- GPS delivery tracking
- Mobile application
- Advanced recommendations
- Loyalty/reward system
- Real-time chat
- Advanced analytics
- Multiple restaurant management
- Advanced notifications

Appendix B – Initial Technology Environment

Component	Technology
Frontend / Web Pages	HTML + CSS
Application / Backend	Asp.Net C#
Database	MySQL
Version Control	Git
Repository	GitHub

Appendix C – Project Lifecycle

The initial project shall follow the following lifecycle:

Project Scenario → Requirements Engineering → Requirements Baseline → System Analysis & Design → Design Baseline → Database Design → Database Baseline → Backend Development → Frontend Development → Integration → Testing → Version 1.0 Release

After the first release:

Change Request → Impact Analysis → Approved Change → Implementation → Testing → New Release

Appendix D – Baseline Statement

Requirements Baseline v1.0

The requirements contained in this document constitute the proposed Version 1.0 requirements baseline for the Online Food Ordering System.

Any modification to these requirements after baseline approval must be controlled through the Software Change Management process.

Baseline Information	Value
Baseline Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____
Approval Signature	_____

Appendix E – Project Document Sequence

The Requirements Baseline is positioned within the overall project as follows:

01 – Initial Client Scenario → 02 – Initial Project Proposal → 03 – Project Plan / Practical Roadmap → 04 – SRS + Requirements Baseline v1.0 → SCM Setup → System Design →

Database Design → Development → Testing → Version 1.0 Release → CR-001 → Impact Analysis → Version 1.1

Final Baseline Note

The Version 1.0 scope has been kept focused on the basic online food ordering process, food item management, customer information, and order management.

Features such as online payment, GPS delivery tracking, mobile applications, advanced recommendations, loyalty systems, and advanced analytics are excluded from the Version 1.0 baseline. These features can later be introduced through formal Change Requests and used to demonstrate the Software Change Management lifecycle.